

Statistical analysis of credit data

Exploratory Data Analysis (EDA) and Statistical Inference

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```
1 library(knitr)
2 options(width=80)
3 knitr::opts_chunk$set(echo = TRUE, message = FALSE, cache = TRUE)
4 opts_chunk$set(tidy.opts=list(width.cutoff=60),tidy=TRUE)
5
6 # packages'
7 library(sys)
8 library(reticulate)
9 library("png")
10 library("rjson")
11 library(imager)
12 library(ggplot2)
13 library(EnvStats)
14 library(kableExtra)
15 library(psych)
16 library(patchwork)
17 library(skimr)
18 library(gt)
19
20
21 library(tidyverse)
22 library(fitdistrplus)
23 library(MASS)
24 library(aqp)
25 library(nortest)
26
27 df<-read.csv("credit_data.csv")
```

0.1 Data

The data set we have been given consists of the following fields: 1. `occupation`: Defines the occupation of the person. 2. `credit_rating`: Is a categorical variable describing how good the persons credit rating is. 3. `number_of_late_payments`: This is the amount of late payments that the person has made. 4. `debt_to_income_ratio`: This is the ratio of you income and your debt. It shows how much of your income has to be payed as debt. 5. `credit_utilization_ratio`: This is the ratio between your total credit limit and your total outstanding balance on all your credit cards.

```
knitr::kable(head(df))
```

id	occupation	credit_rating	number_of_late_payments	debt_to_income_ratio	credit_utilization_ratio
1	Engineer	Fair	5	0.27	0.41
2	Clerk	Fair	4	0.43	0.39
3	Manager	Fair	1	0.40	0.35
4	Nurse	Fair	2	0.30	0.52
5	Clerk	Poor	4	0.66	0.51
6	Salesperson	Good	9	0.19	0.20

```
1 import numpy as np
2 import matplotlib
3 import matplotlib.pyplot as plt
4 import scipy.stats as sc
5 import seaborn as sb
6 import statistics as stts
7 import pandas as pd
8 from fitter import Fitter, get_common_distributions, get_distributions
9 import csv
10 import statsmodels.api as sm
11 import json
12 import math
13 from distfit import distfit
14
15 id=[]
16 occupation=[]
17 credit_rating=[]
18 number_of_late_payments=[]
19 debt_to_income_ratio=[]
20 credit_utilization_ratio=[]
```

```

21 alldata=dict
22
23 allModels=['Fair','Good','Poor']
24 alloccupations= [
25     "Engineer",
26     "Clerk",
27     "Manager",
28     "Nurse",
29     "Salesperson",
30     "Teacher"]
31
32 allcolors="red","green","blue","yellow","orange","black"
33 with open('credit_data.csv', newline='') as csvfile:
34     reader = csv.DictReader(csvfile)
35
36     for row in reader:
37         id.append(float(row['id']))
38         occupation.append(row['occupation'])
39         credit_rating.append(row['credit_rating'])
40         number_of_late_payments.append(float(row['number_of_late_payments']))
41         debt_to_income_ratio.append(float(row['debt_to_income_ratio']))
42         credit_utilization_ratio.append(float(row['credit_utilization_ratio']))
43
44
45     def generateMeanStandardDev(thisList):
46
47         tmean=stts.mean(thisList)
48         tstdev=stts.stdev(thisList)
49         tvar=stts.variance(thisList)
50         tharmmean=stts.harmonic_mean(thisList)
51         tskev= sc.skew(thisList,axis=0,bias=True)
52         tkurt=sc.kurtosis(thisList,axis=0,bias=True)
53         tgeomean=0
54         if 0 in thisList:
55             tgeomean=0
56         else:
57             tgeomean=stts.geometric_mean(thisList)
58         #print(tmean,tstdev,tvar)
59         return [tmean,tstdev,tvar,tharmmean,tgeomean,tskev,tkurt,]
60
61     def eachModelthing(row,query):
62         icount=1

```

```

63     allModels=['Fair','Good','Poor']
64     for x in allModels:
65         #print(x)
66
67         thisoccupation={'id':[],'occupation':[],'credit_rating':[],'number_of_late
68     for thisid in id:
69         thisid=int(thisid)-1
70         if credit_rating[thisid]==x :
71
72             thisoccupation['id'].append(1)
73             thisoccupation['occupation'].append(occupation[thisid])
74             thisoccupation['credit_rating'].append(credit_rating[thisid])
75             thisoccupation['sum_of_each_rating']
76             thisoccupation['number_of_late_payments'].append(float(number_of_l
77             thisoccupation['debt_to_income_ratio'].append(float(debt_to_income
78             thisoccupation['credit_utilization_ratio'].append(float(credit_utili
79
80
81     data = {
82         "credit_utilization_ratio": thisoccupation['credit_utilization_ratio'],
83         "number_of_late_payments": thisoccupation['number_of_late_payments'],
84         "debt_to_income_ratio":thisoccupation['debt_to_income_ratio'],
85         "occupation":thisoccupation['occupation'],
86         "credit_rating":thisoccupation['credit_rating'],
87         "id":thisoccupation['id']
88     }
89     df = pd.DataFrame(data)
90     df.head()
91
92     sb.kdeplot(data=df, x=query, multiple="stack",color=allcolors[row],label=x
93     icount+=1
94
95 def ShowDistribution(collum_name,place):
96     data = {
97         "credit_utilization_ratio": credit_utilization_ratio,
98         "number_of_late_payments": number_of_late_payments,
99         "debt_to_income_ratio":debt_to_income_ratio,
100        "occupation":occupation,
101        "credit_rating":credit_rating,
102        "id":id
103    }
104    df = pd.DataFrame(data)

```

```

104         df.head()
105         thisfield = df[collum_name].values
106         stuff=generateMeanStandardDev(data[collum_name])
107         mean=stuff[0]
108         stdev=stuff[1]
109         vari=stuff[2]
110         harmmean=stuff[3]
111         geomean=stuff[4]
112         linewidth=1
113         gx=sb.kdeplot(data=df, x=collum_name, multiple="stack",ax=place,label=collum_name)
114         gx.legend(loc="upper right")
115         gx.set(xlabel=None)
116         gx.set(ylabel=None)
117         place.axvline(x = mean, color="red" , linewidth =linewidth)
118         if harmmean !=0 :
119             place.axvline(x = harmmean, color="black" , linewidth =linewidth)
120         if geomean !=0 :
121             place.axvline(x = geomean, color="white" , linewidth =linewidth)
122         place.axvline(x = mean-vari ,color="green",linewidth =linewidth)
123
124         place.axvline(x =mean+vari, color="green",linewidth =linewidth )
125
126         place.axvline(x =mean-stdev, color="yellow" , linewidth =linewidth )
127         place.axvline(x =mean+stdev, color="yellow" ,linewidth =linewidth )
128         jdataDict={
129             "name":collum_name,
130             "mean":stuff[0],
131             "standarddev":stuff[1],
132             "variance":stuff[2],
133             "harmmean":stuff[3],
134             "geomean":stuff[4],
135             "skewness":stuff[5],
136             "kurtosis":stuff[6]
137         }
138         return jdataDict
139
140     def boxplots(thing1_name):
141         data = {
142             "credit_utilization_ratio": credit_utilization_ratio,
143             "number_of_late_payments": number_of_late_payments,
144             "debt_to_income_ratio":debt_to_income_ratio,

```

```

145         "occupation":occupation,
146         "credit_rating":credit_rating,
147         "id":id
148     }
149     df = pd.DataFrame(data)
150     df.head()
151     sb.boxplot(df,y=thing1_name,x="credit_rating",hue="occupation").legend( loc='best'
152 def compare2things(thing1_name,thing2_name,df):
153
154
155     regression=(sc.linregress(x=data[thing1_name],y=data[thing2_name]))
156     #print(regression)
157     b=regression[0]
158     a=regression[1]
159     point1=a+b*min(data[thing1_name])
160     point2=a+b*max(data[thing1_name])
161
162     pearsonC=sc.pearsonr(data[thing1_name],data[thing2_name])
163
164     txtreturn="name1 "+thing1_name+"\nname2"+thing2_name+"\npearson_coefficient"+str(p
165     jdataDict={
166         "name1":thing1_name,
167         "name2":thing2_name,
168         "pearson_coefficient":pearsonC[0],
169         "pearson_p_value":pearsonC[1],
170         "regression_line":f"y={a}+{b}*x"
171     }
172     return jdataDict
173     #return txtreturn
174
175
176
177
178     #print(len(credit_utilization_ratio))
179     #print(len(number_of_late_payments))
180     #print(len(debt_to_income_ratio))
181
182     #print(len(occupation))
183
184     #print(len(credit_rating))
185

```

```

186 #print(len(id))
187 data = {
188     "credit_utilization_ratio": credit_utilization_ratio,
189     "number_of_late_payments": number_of_late_payments,
190     "debt_to_income_ratio":debt_to_income_ratio,
191     "occupation":occupation,
192     "credit_rating":credit_rating,
193     "id":id
194 }
195 df = pd.DataFrame(data)
196 df.head()
197
198 '''
199 incomedist=ShowDistribution("debt_to_income_ratio" )
200 latedist=ShowDistribution("number_of_late_payments" )
201 creditdist=ShowDistribution("credit_utilization_ratio")
202 '''
203 f, axes = plt.subplots(3)
204 f. set_size_inches((8, 6))
205 incomedist=ShowDistribution("debt_to_income_ratio",place=axes[0] )
206 latedist=ShowDistribution("number_of_late_payments",place=axes[1] )
207 creditdist=ShowDistribution("credit_utilization_ratio",place=axes[2] )
208 plt.savefig("Distributions1.png")
209
210
211
212 c1=compare2things("debt_to_income_ratio","number_of_late_payments",df)
213 c2=compare2things("credit_utilization_ratio","debt_to_income_ratio",df)
214 c3=compare2things("number_of_late_payments","credit_utilization_ratio",df)
215
216 f = open("c1.txt", "w")
217 f.write( json.dumps(c1,indent=4,))
218 f.close()
219 f = open("c2.txt", "w")
220 f.write( json.dumps(c2,indent=4,))
221 f.close()
222 f = open("c3.txt", "w")
223 f.write( json.dumps(c3,indent=4,))
224 f.close()
225
226 plt.tight_layout()

```

```

227 f, axes = plt.subplots(1)
228 f. set_size_inches((10, 5))
229 boxplots("debt_to_income_ratio")
230 plt.savefig("BoxC.png")
231 plt.tight_layout()
232 f, axes = plt.subplots(1)
233 f. set_size_inches((10, 5))
234 boxplots("credit_utilization_ratio")
235 plt.savefig("BoxM.png")
236 plt.tight_layout()
237 f, axes = plt.subplots(1)
238 f. set_size_inches((10, 5))
239 boxplots("number_of_late_payments")
240 plt.savefig("BoxW.png")
241
242
243 #####
244 #plt.show()
245 out={
246     "Probability_Distributions":{
247         "incomedist":incomedist,
248         "latedist":latedist,
249         "creditdist":creditdist}
250 }
251 jsout=json.dumps(out,indent=4,)
252 #put json into a text file:
253 txtComparisons=out["Probability_Distributions"]
254 global txtout
255 txtout=""
256 for l in txtComparisons:
257
258     for x in txtComparisons[l]:
259         #print(x + " \t"+str(txtComparisons[l][x]))
260         num=len(x)
261         strx=x+"="*(20-num)
262         if type(txtComparisons[l][x])!=str:
263             txtout+=(strx+str(round(float(txtComparisons[l][x]),4)))
264         else:
265             txtout+=(strx +str(txtComparisons[l][x]))
266         txtout+="\n"
267

```



```

268         txtout+="\n"
269
270     f = open("numbers.txt", "w")
271     f.write(txtout)
272     f.close()

```

0.2 EDA

The goal of this EDA is to understand “credit_data.csv”’s underlying structure, find correlations, test hypotheses, fit confidence intervals and fit density distributions.

```

1 f = open("numbers.txt", "r").read()
2 print(f)

```

```

name=====debt_to_income_ratio
mean=====0.3355
standarddev=====0.1145
variance=====0.0131
harmmean=====0.3032
geomean=====0.3183
skewness=====1.1259
kurtosis=====1.2089

name=====number_of_late_payments
mean=====3.2652
standarddev=====2.6057
variance=====6.7896
harmmean=====0.0
geomean=====0.0
skewness=====1.0957
kurtosis=====1.0068

name=====credit_utilization_ratio
mean=====0.4096
standarddev=====0.1064
variance=====0.0113
harmmean=====0.378
geomean=====0.3947
skewness=====0.0917
kurtosis===== -0.0576

```

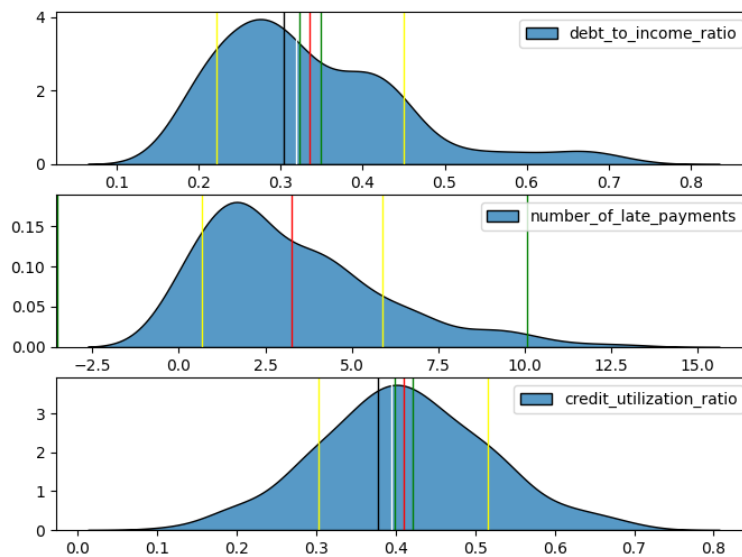
The skewness of the fields indicates if the data will be negatively or positively skewed. If the value is between -1 and -0.5: the data is negatively skewed. If it is between 0.5 to 1: the data is positively skewed. The kurtosis of the data indicated the density of the tail ends of the density plot.

The table below shows some of the above data in density tables for the respective fields:

- mean -> red
- harmonic mean -> black
- geometric mean -> green
- variance about the mean -> green
- standard deviation about the mean -> yellow

* note that not all of the fields have a harmonic or geometric mean because of the way that they are calculated if there is a 0 in the data set then the harmonic and geometric mean will not exist.

```
1 im <- load.image("Distributions1.png")
2 par(mar = c(0, 0, 0, 0))
3 plot(im, axes=FALSE)
```



In the table above you can visually confirm the data given in the EDA: Such as the skewness of late payments and debt ratio. It should be noted that none of the number of late payments

actually has a negative value, it is simply a result of the graphing technique that is smoothing the density distribution to be approximate.

This table shows each 1/4th percentile for :

- credit_utilization_ratio
- number_of_late_payments
- debt_to_income_ratio

```

1 skimmedDFinitial <- df %>%
2   skim() %>%
3   yank("numeric")
4 skimmedDF <- skim(df) %>%
5   dplyr::select(skim_variable, numeric.p25,numeric.p50, numeric.p75)
6 skimmedDF<-skimmedDF[4:6,]
7 kable(skimmedDF, digits = 2, align = "c") %>%
8   kable_styling(
9     full_width = FALSE,
10    font_size = 11,
11    latex_options = c("striped", "hold_position")
12  ) %>%
13  row_spec(0,font_size = 16, color = "white", background = "#2d9103") %>%
14  row_spec(1:nrow(skimmedDF), color = "black", background = scales::seq_gradient_pal("green", "red"))
15  column_spec(1, bold = TRUE, color = "black", background = "white") %>%
16  add_header_above(c("Percentile's of each field" = ncol(skimmedDF)))

```

Percentile's of each field			
skim_variable	numeric.p25	numeric.p50	numeric.p75
number_of_late_payments	1.00	2.00	5.00
debt_to_income_ratio	0.26	0.31	0.40
credit_utilization_ratio	0.34	0.41	0.48

These percentiles are then used to get the outliers of the data. Outliers are a good way of detecting and accounting for anomalies when conducting further tests on data.

```

1 Q1m <- quantile(df$credit_utilization_ratio, .25)
2 Q3m <- quantile(df$credit_utilization_ratio, .75)
3 IQRm <- IQR(df$credit_utilization_ratio)
4
5 Q1w <- quantile(df$number_of_late_payments, .25)
6 Q3w <- quantile(df$number_of_late_payments, .75)

```

```

7   IQRw <- IQR(df$number_of_late_payments)
8
9   Q1s <- quantile(df$debt_to_income_ratio, .25)
10  Q3s <- quantile(df$debt_to_income_ratio, .75)
11  IQRs <- IQR(df$debt_to_income_ratio)
12
13
14  outliers <- subset(df,
15                    df$credit_utilization_ratio<(Q1m - 1.5*IQRm) | df$credit_utilization_ratio>(Q3m + 1.5*IQRm)
16                    df$number_of_late_payments<(Q1w - 1.5*IQRw) | df$number_of_late_payments>(Q3w + 1.5*IQRw)
17                    df$debt_to_income_ratio<(Q1s - 1.5*IQRs) | df$debt_to_income_ratio>(Q3s + 1.5*IQRs)
18                    )
19  o1<- subset(df,df$credit_utilization_ratio<(Q1m - 1.5*IQRm) | df$credit_utilization_ratio>(Q3m + 1.5*IQRm)
20  o2<- subset(df,df$number_of_late_payments<(Q1w - 1.5*IQRw) | df$number_of_late_payments>(Q3w + 1.5*IQRw)
21  o3<- subset(df,df$debt_to_income_ratio<(Q1s - 1.5*IQRs) | df$debt_to_income_ratio>(Q3s + 1.5*IQRs)
22  outliers<-outliers[,]
23  kable(outliers, digits = 0.02, align = "c") %>%
24    kable_styling(
25      full_width = FALSE,
26      font_size = 11,
27      latex_options = c("striped", "hold_position")
28    ) %>%
29    row_spec(0,font_size = 16, color = "white", background = "#2d9103")%>%
30    column_spec(1, bold = TRUE, color = "black", background = "white") %>%
31    add_header_above(c("Table of outliers" = ncol(outliers)+1))

```

Table of outliers

	id	occupation	credit_rating	number_of_late_payments	debt_to_income_ratio
5	5	Clerk	Poor	4	1
17	17	Teacher	Poor	7	0
18	18	Engineer	Fair	1	0
36	36	Engineer	Fair	2	1
40	40	Clerk	Poor	4	1
42	42	Salesperson	Poor	3	1
76	76	Salesperson	Fair	13	0
77	77	Teacher	Fair	5	1
91	91	Engineer	Fair	1	1
106	106	Nurse	Poor	3	1
158	158	Engineer	Poor	6	1
201	201	Teacher	Poor	10	1

```

1  my_df <- data.frame(
2    d = "Outliers",
3    credit_utilization_ratio = nrow(o1),
4    number_of_late_payments = nrow(o2),
5    debt_to_income_ratio = nrow(o3)
6  )
7  colnames(my_df)[1] <- ""
8
9  # Create a kable table
10 kable(my_df, align = "c")%>%
11   kable_styling(
12     full_width = FALSE,
13     font_size = 11,
14     latex_options = c("striped", "hold_position")
15   ) %>%
16   row_spec(0,font_size = 16, color = "white", background = "#2d9103") %>%
17   column_spec(1, bold = TRUE, color = "black", background = "white")%>%
18   add_header_above(c("Number of outliers per independant variable" = ncol(my_df)))

```

Number of outliers per independant variable			
	credit_utilization_ratio	number_of_late_payments	debt_to_income_ratio
Outliers	2	2	9

Since there are only 2 outliers in the credit utilization ratio and number of late payments columns, they probably will not impact any of the upcoming analysis. There is however a high amount of outliers in the debt to income ratio column. this can be confirmed in the high skewness and kurtosis of debt_to_income_ratio. It is still important to include outliers in data analysis unless there are clear reasons to exclude them. But if any inconsistencies appear in the results, it may be worthwhile to check if the outliers are causing these issues. It should also be noted from the table of outliers that only “Poor” and “Fair” credit ratings are in the outliers.

0.3 Compring Data

0.3.1 Checking regression lines

This section compares certain fields with each other to see if they have any significant statistical correlation. the fields being compared are:

- (number_of_late_payments,credit_utilization_ratio)
- (credit_utilization_ratio,debt_to_income_ratio)
- (debt_to_income_ratio,number_of_late_payments)

The Pearson correlation coefficient:

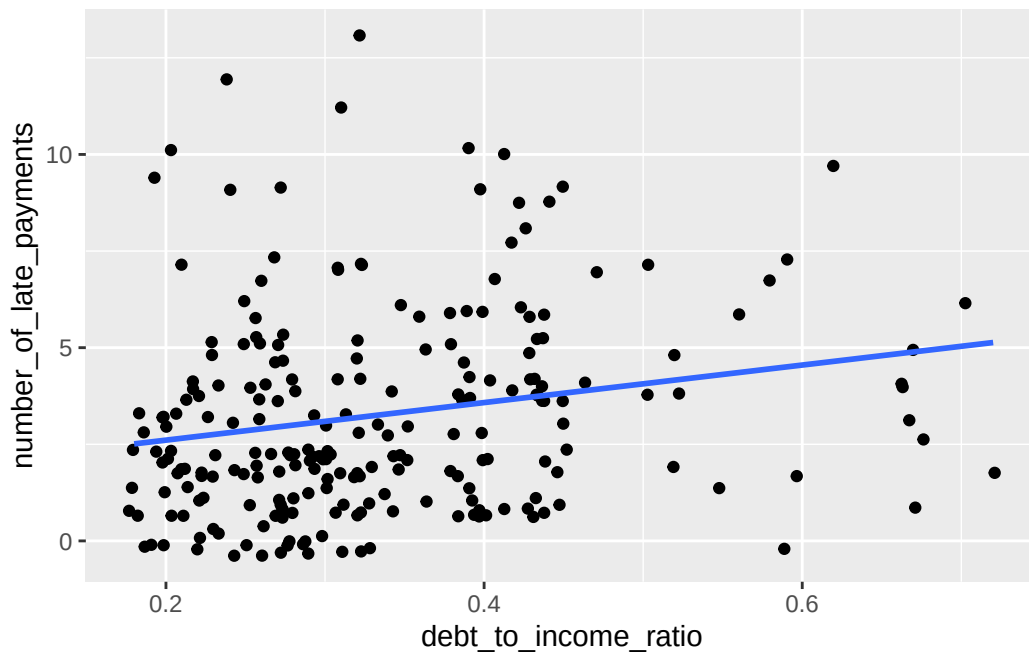
- A value close to 1 indicates a strong positive linear relationship.
- A value close to -1 indicates a strong negative linear relationship.
- A value close to 0 suggests there is little to no linear relationship between the variables.

P-value:

- A p-value below a common limit such as 0.05 would indicate that the observed correlation is statistically significant.
- A p-value higher than 0.05 suggests that there is no statistical significant correlation between the variables.

0.3.1.1 number_of_late_payments VS debt_to_income_ratio

```
ggplot(df,aes(x=debt_to_income_ratio,y=number_of_late_payments))+
  geom_jitter()+geom_smooth(method="lm",se=FALSE)+
  xlab("debt_to_income_ratio")+ylab("number_of_late_payments")
```



```

1 f = open("c1.txt", "r").read()
2 print(f)

```

```

{
  "name1": "debt_to_income_ratio",
  "name2": "number_of_late_payments",
  "pearson_coefficient": 0.21345484679247567,
  "pearson_p_value": 0.0011252254123582809,
  "regression_line": "y=1.6358238247930685+4.856927427392357*x"
}

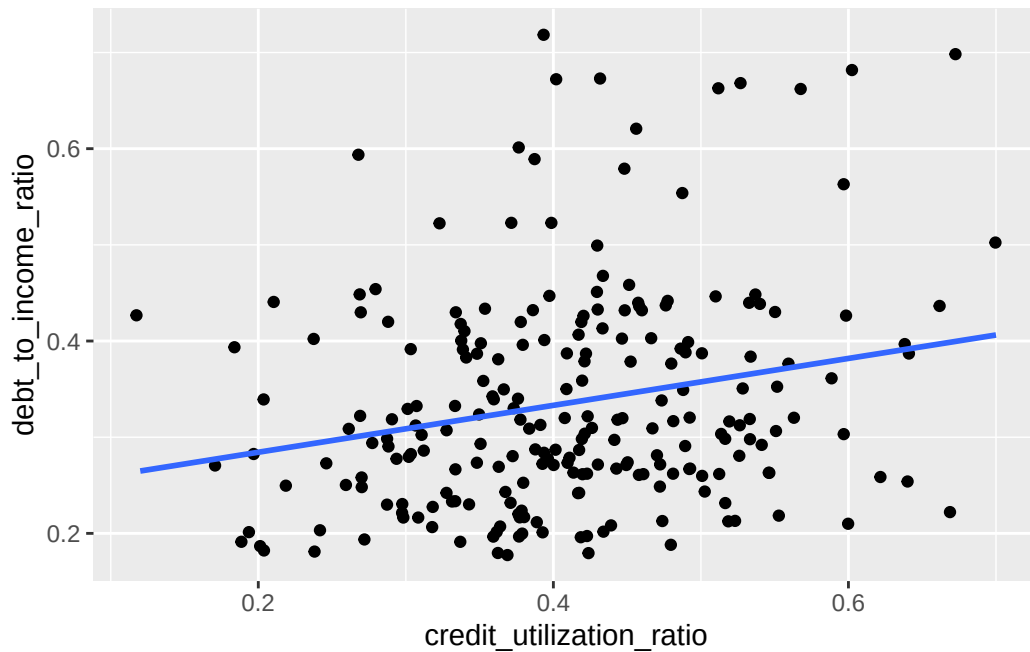
```

0.3.1.2 credit_utilization_ratio VS debt_to_income_ratio

```

1 ggplot(df,aes(x=credit_utilization_ratio,y=debt_to_income_ratio))+
2   geom_jitter()+geom_smooth(method="lm",se=FALSE)+
3   xlab("credit_utilization_ratio")+ylab("debt_to_income_ratio")

```



```

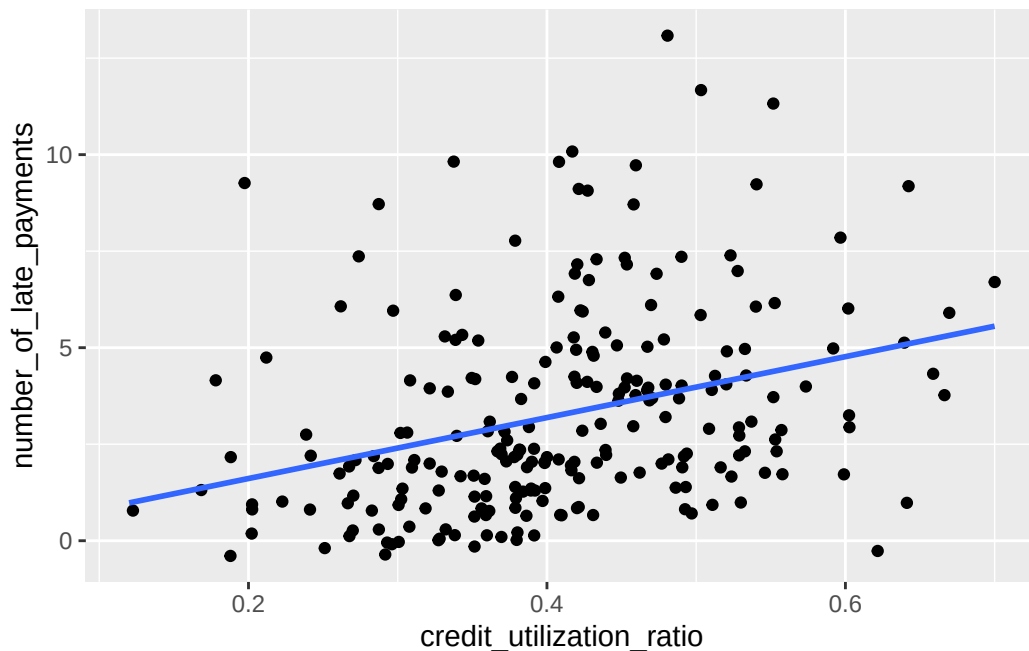
1 f = open("c2.txt", "r").read()
2 print(f)

```

```
{
  "name1": "credit_utilization_ratio",
  "name2": "debt_to_income_ratio",
  "pearson_coefficient": 0.22665626214920928,
  "pearson_p_value": 0.0005324569590632021,
  "regression_line": "y=0.23559770056582663+0.24384384746693427*x"
}
```

0.3.1.3 number_of_late_payments VS credit_utilization_ratio

```
1 ggplot(df,aes(x=credit_utilization_ratio,y=number_of_late_payments))+
2   geom_jitter()+geom_smooth(method="lm",se=FALSE)+
3   xlab("credit_utilization_ratio")+ylab("number_of_late_payments")
```



```
1 f = open("c3.txt", "r").read()
2 print(f)
```

```
{
  "name1": "number_of_late_payments",
  "name2": "credit_utilization_ratio",
  "pearson_coefficient": 0.3223435625713474,
```



```

    "pearson_p_value": 5.854980429006792e-07,
    "regression_line": "y=0.36661236154571053+0.013167985145787715*x"
}

```

0.3.1.4 Interpretation of regression lines

Here we can see that there are no strong linear correlations between the data sets. This is because all the pearson p-values are higher than 0.05 which suggests that there is no statistical significant correlation between the above data sets.

0.3.2 Comparing fields by region

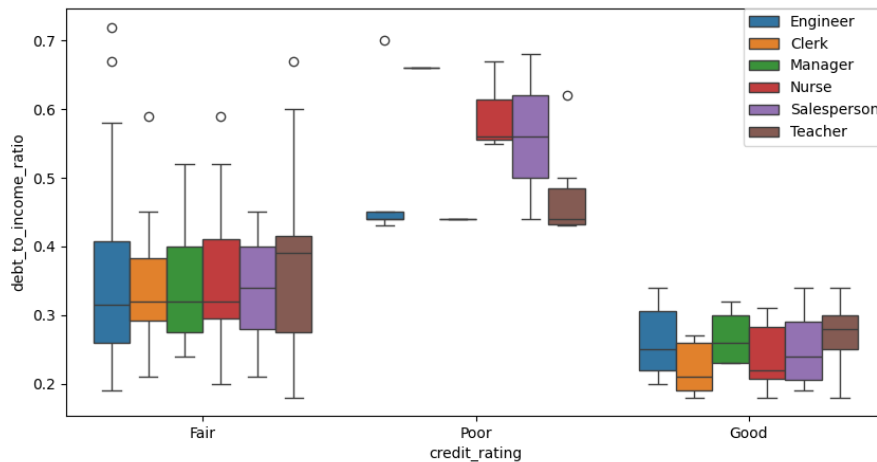
Below are the visual comparisons of each type of credit rating using box plots

0.3.2.1 debt to income ratio

```

1 im <- load.image("BoxC.png")
2 par(mar = c(0, 0, 0, 0))
3 plot(im, axes=FALSE)

```

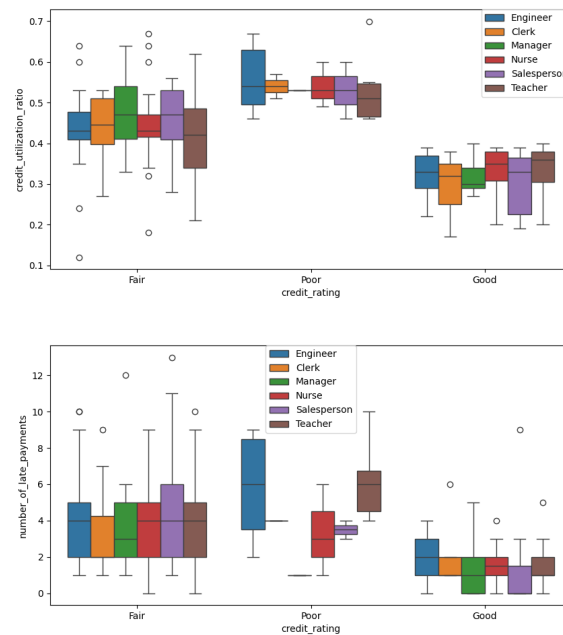


From this box plot it is clear that credit rating influences the debt to income ratio. Poor credit rating having the highest debt to income ratio, good have the lowest debt to income ratio and

fair lie between them. Comparing the other two numeric independent variables with credit rating follows a similar trend:

0.3.2.2 Number of late payments and credit utilization ratio

```
1 par(mfrow = c(2, 1))
2
3 im <- load.image("BoxM.png")
4 par(mar = c(0, 0, 0, 0))
5 plot(im, axes=FALSE)
6
7 im <- load.image("BoxW.png")
8 par(mar = c(0, 0, 0, 0))
9 plot(im, axes=FALSE)
```



From the box plots of warranty claims and credit utilization ratio one can see that the credit rating does not impact the warranty claims or credit utilization ratio as much as it effects debt to income ratio. The strange observation of poor credit rating can be attributed to high variance within the poor credit rating subset of the data.

0.4 Statistical Inference

This section is for statistical inference.

0.4.1 Estimation

0.4.1.1 Goodness-of-fit Test

Using a Shapiro-wilk test to do a goodness of fit test on the credit_utilization_ratio The test will check Normal, Gamma and Lognormal distributions.

```
1 chosenDist<-EnvStats::distChoose(y=df$credit_utilization_ratio)
2 chosenDist
```

Results of Choosing Distribution

Candidate Distributions:	Normal Gamma Lognormal
--------------------------	------------------------------

Choice Method:	Shapiro-Wilk
----------------	--------------

Type I Error per Test:	0.05
------------------------	------

Decision:	Normal
-----------	--------

Estimated Parameter(s):	mean = 0.4096087 sd = 0.1064444
-------------------------	------------------------------------

Estimation Method:	mvue
--------------------	------

Data:	df\$credit_utilization_ratio
-------	------------------------------

Sample Size:	230
--------------	-----

Test Results:

Normal	
Test Statistic:	W = 0.9954637
P-value:	0.7329635

Gamma

Test Statistic:	W = 0.9831784
P-value:	0.007872058

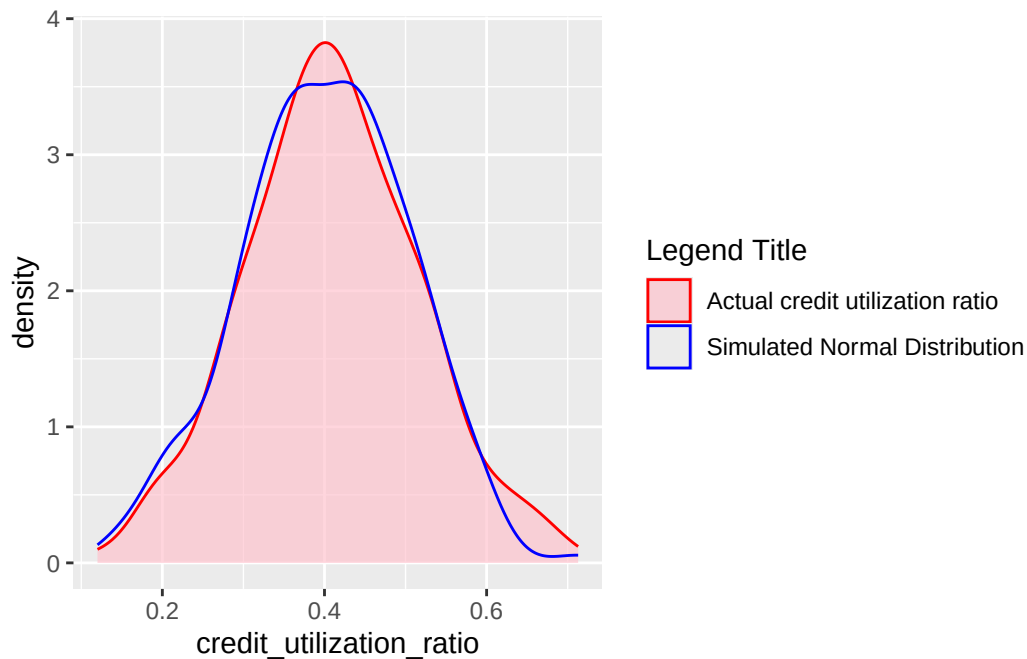
Lognormal

Test Statistic:	W = 0.9625792
P-value:	9.888096e-06

From this test we can see that it is most likely that the credit utilization ratio has a normal distribution as that has the highest test statistic and p value.

This graph shows the density distribution of credit utilization ratio and the expected density based off of the mean, standard deviation and variance of the sample data.

```
1 ggplot(df, aes(x = credit_utilization_ratio)) +  
2   geom_density(aes(colour = "Actual credit utilization ratio"), fill = "pink", alpha = 0.6)  
3  
4   geom_density(aes(x = rnorm(n = length(credit_utilization_ratio),  
5                       mean = mean(credit_utilization_ratio),  
6                       sd = sd(credit_utilization_ratio)),  
7               colour = "Simulated Normal Distribution"), fill = "blue", alpha = 0.0)  
8  
9   scale_colour_manual(values = c("Actual credit utilization ratio" = "red",  
10                                "Simulated Normal Distribution" = "blue")) +  
11  
12  guides(colour = guide_legend(title = "Legend Title"))
```



This provides a visual confirmation that credit utilization ratio has a normal distribution.

0.4.2 Interval Estimation

Density Plot for credit utilization ratio, with Shaded Critical Regions of alpha value 0.05. This graph of the credit utilization ratio shows the rejection regions using the critical values.

```
1 alpha <- 0.05
2 critical_value_left <- qnorm(alpha / 2, mean = mean(df$credit_utilization_ratio), sd = sd(df$credit_utilization_ratio))
3 critical_value_right <- qnorm(1 - alpha / 2, mean = mean(df$credit_utilization_ratio), sd = sd(df$credit_utilization_ratio))
4 cat(
5   paste(" lower bound :", critical_value_left, sep=" "),"\n",
6   paste("upper bound :", critical_value_right, sep=" ")
7 )
```

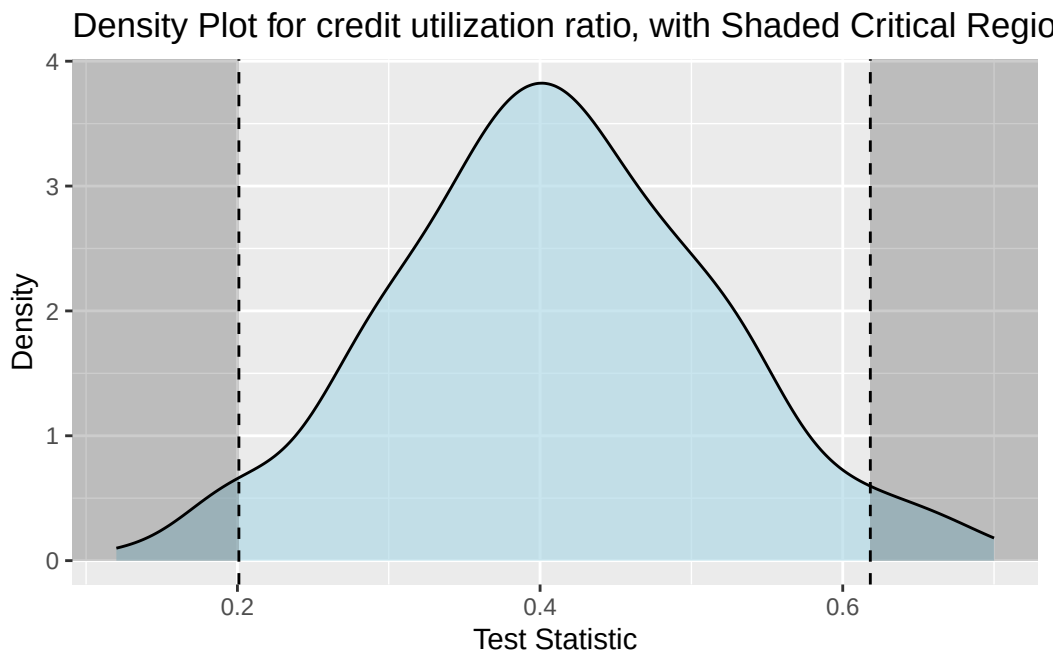
```
lower bound : 0.200981472092954
upper bound : 0.618235919211394
```

```
1 ggplot(df, aes(x = credit_utilization_ratio)) +
2   geom_density(fill = "lightblue", alpha = 0.65) +
3   geom_vline(xintercept = c(critical_value_left, critical_value_right),
```

```

4         linetype = "dashed", color = "black") +
5     annotate("rect", xmin = -Inf, xmax = critical_value_left, ymin = 0, ymax = Inf,
6             alpha = 0.2, fill = "black") +
7     annotate("rect", xmin = critical_value_right, xmax = Inf, ymin = 0, ymax = Inf,
8             alpha = 0.2, fill = "black") +
9     labs(title = "Density Plot for credit utilization ratio, with Shaded Critical Regions of

```



0.4.3 Hypothesis testing

```

1 a <- EnvStats::varTest(x=df$credit_utilization_ratio,conf.level = 0.05)
2 a$method

```

```
[1] "Chi-Squared Test on Variance"
```

```
1 a
```

Results of Hypothesis Test

Null Hypothesis:	variance = 1
Alternative Hypothesis:	True variance is not equal to 1
Test Name:	Chi-Squared Test on Variance
Estimated Parameter(s):	variance = 0.01133041
Data:	df\$credit_utilization_ratio
Test Statistic:	Chi-Squared = 2.594665
Test Statistic Parameter:	df = 229
P-value:	1.784685e-175
5% Confidence Interval:	LCL = 0.01129707 UCL = 0.01143041

With this we can see that the p-value is much less than the test statistic. therefore we reject the null hypothesis, meaning that we have strong statistical evidence to say that the actual variance of credit_utilization_ratio is not equal to 1.

0.5 Insights and Conclusion

Jump to outliers Since there were not any inconsistencies in analysis of the data while the outliers were included it is safe to assume that they don't significantly impact the analysis. However it may be beneficial to redo some of the tests without the outliers if we want to do a more indepth analysis in the future. Jump to regression Lines It was found from the pearson p-values of the that there are no significant correlations between the credit_utilization_ratio, credit_utilization_ratio and customer_satisfaction_score. this means that these fields do not affect each other in any way when categorised as a whole set. However they are linked to credit rating as we will see in the box plot analysis.

Jump to Box plots Comparing the different credit ratings: When visually comparing the credit_utilization_ratio, credit_utilization_ratio and customer_satisfaction_score based on credit rating one can see that.

"Poor" has the highest debt to income ratio, "Good" has the lowest debt to income ratio and "Fair" lie between them.

This same trend can be observed with slightly less extremity when looking at credit utilisation ratio and number of late payments.

The strange artefacts in “Poor” section of the box plots is attributed to a small sample size and the values in “Poor” category being grouped closely and spread widely. An example of this is “Poor” Engineers having a wide spread in number of late payments and credit utilisation rate, but being grouped close together in debt to income ratio.

Interesting observation

The lack of correlation means that our independent variables have a weak colinearity. The distinct change of spread of each independent variable when categorised in the box plots is important to note here as it means that this dataset can be used for multivariable regression or even machine learning to predict someones credit rating given their occupation, number_of_late_payments, debt_to_income_ratio, credit_utilization_ratio. To confirm this suspicion an anova test can be conducted:

```
1 anova_late_payments <- aov(number_of_late_payments ~ credit_rating, data = df)
2 summary(anova_late_payments)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
credit_rating	2	325.4	162.71	30.04	2.66e-12 ***
Residuals	227	1229.4	5.42		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
1 anova_dti <- aov(debt_to_income_ratio ~ credit_rating, data = df)
2 summary(anova_dti)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
credit_rating	2	1.214	0.6069	76.99	<2e-16 ***
Residuals	227	1.789	0.0079		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
1 anova_utilization <- aov(credit_utilization_ratio ~ credit_rating, data = df)
2 summary(anova_utilization)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
credit_rating	2	1.109	0.5545	84.72	<2e-16 ***


```
Residuals      227  1.486  0.0065
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

From this we see that all the $\Pr(>F)$ values are less than 0.0001. this means that there are significant differences between credit rating subgroups. This confirms the above assumption that the data can be used in multivariable regression and machine learning to predict credit ratings.

0.5.1 Recommendations

Based on the box plots and the anova test: if someone has a bad credit rating and would like to improve it, they should: lower their credit utilisation ratio(spend less of your total available credit), and they should make less late payments. They must also most importantly reduce their debt to income ratio.